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(54) **LIFTING DEVICE AND SHADING DEVICE FOR SUNSHADES**

(71) Applicant: **HANGZHOU XIAOYI INNOVATIONS TECHNOLOGY CO., LTD.**, Hangzhou (CN)

(72) Inventors: **Ruohan Mao**, Hangzhou (CN); **Wenjie Zhang**, Hangzhou (CN); **Rongzhe Liang**, Hangzhou (CN); **Yunwen Kong**, Hangzhou (CN)

(73) Assignee: **Ningbo Esoul digital Technology Co., LTD.**, Ningbo (CN)

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CPC ***A45B 25/143*** (2013.01); ***A45B 23/00*** (2013.01); *A45B 2023/0012* (2013.01)

(58) **Field of Classification Search**
CPC A45B 23/00; A45B 25/143; A45B 2023/0012
See application file for complete search history.

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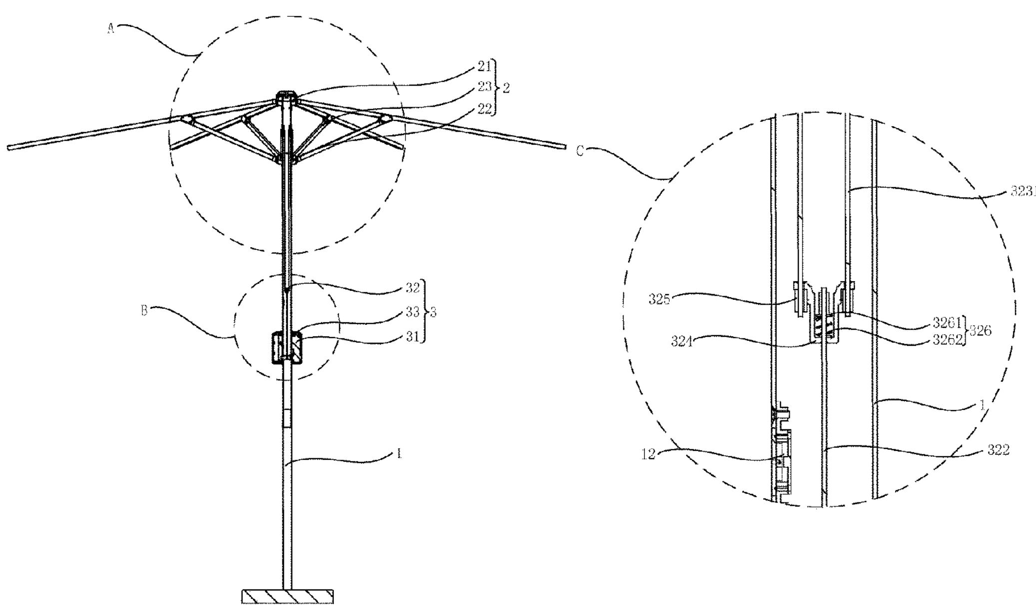
Primary Examiner — Noah Chandler Hawk
(74) *Attorney, Agent, or Firm* — Nitin Kaushik

(57) **ABSTRACT**

The present utility model provides a lifting device and shading device for a sunshade, which comprises a mounting bracket, a sunshade cloth, a support mechanism and a lifting device set on the mounting bracket, wherein the support mechanism comprises a fixed support member arranged at the top of the mounting bracket, a movable support member sliding on the mounting bracket, and a support component connected to the fixed support member and the movable support member respectively and used for unfolding or closing the sunshade cloth, and the lifting device comprises a driving motor, a transmission mechanism connected to the driving motor and used to drive the lifting motion of the movable support member relative to the mounting bracket, and a control component used to control the start and stop of the driving motor.

7 Claims, 5 Drawing Sheets



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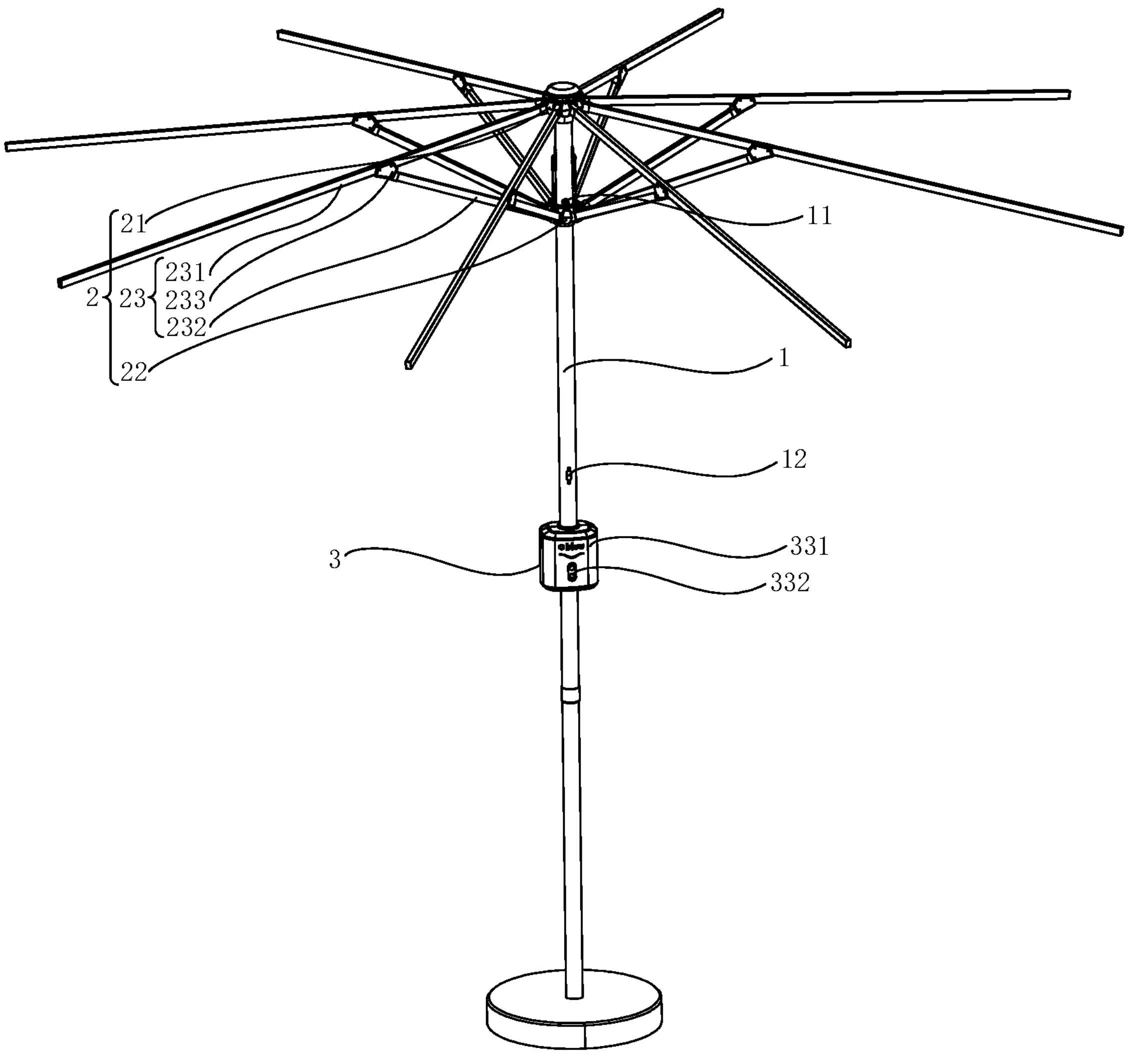

FIG. 1

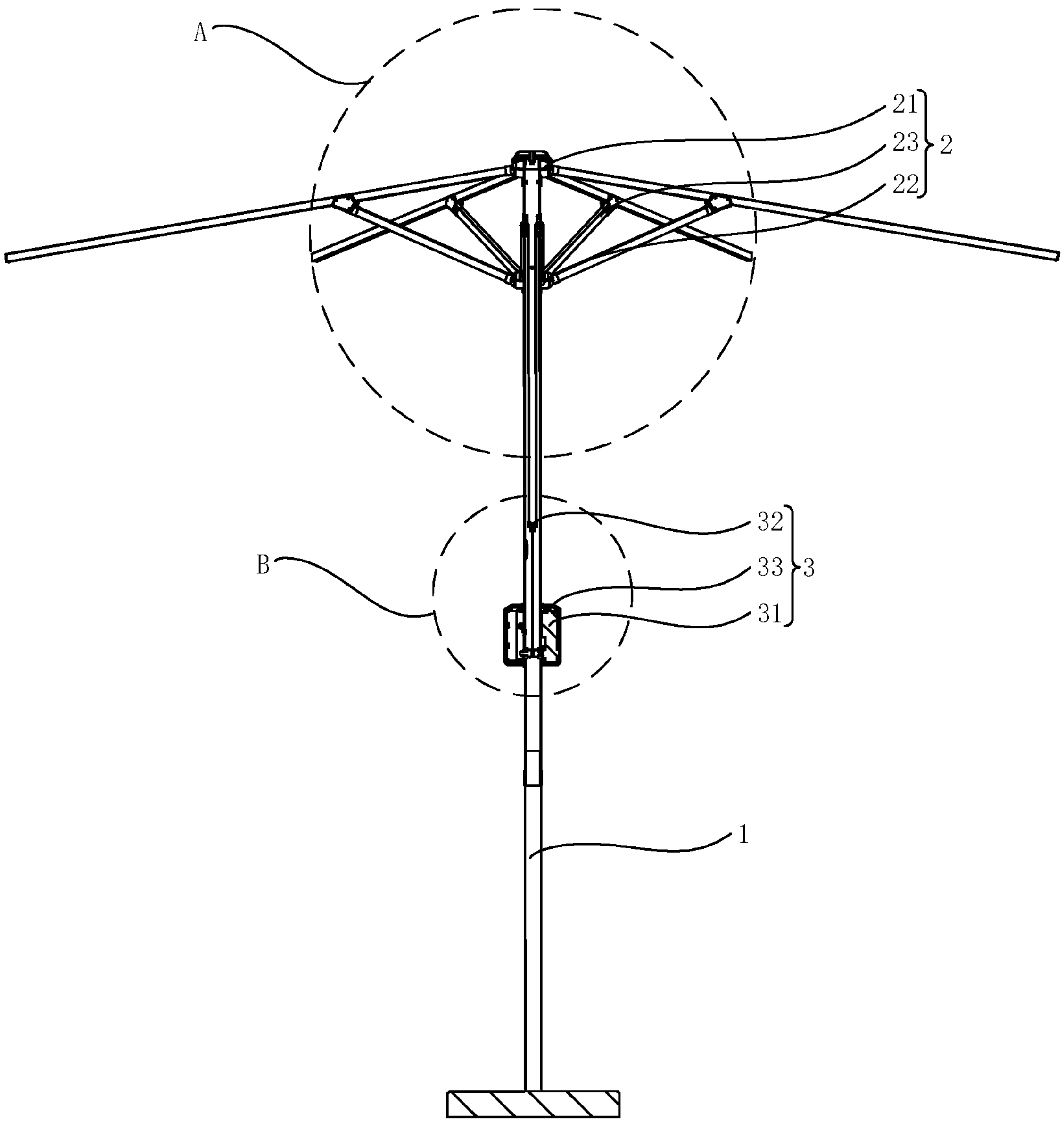

FIG. 2

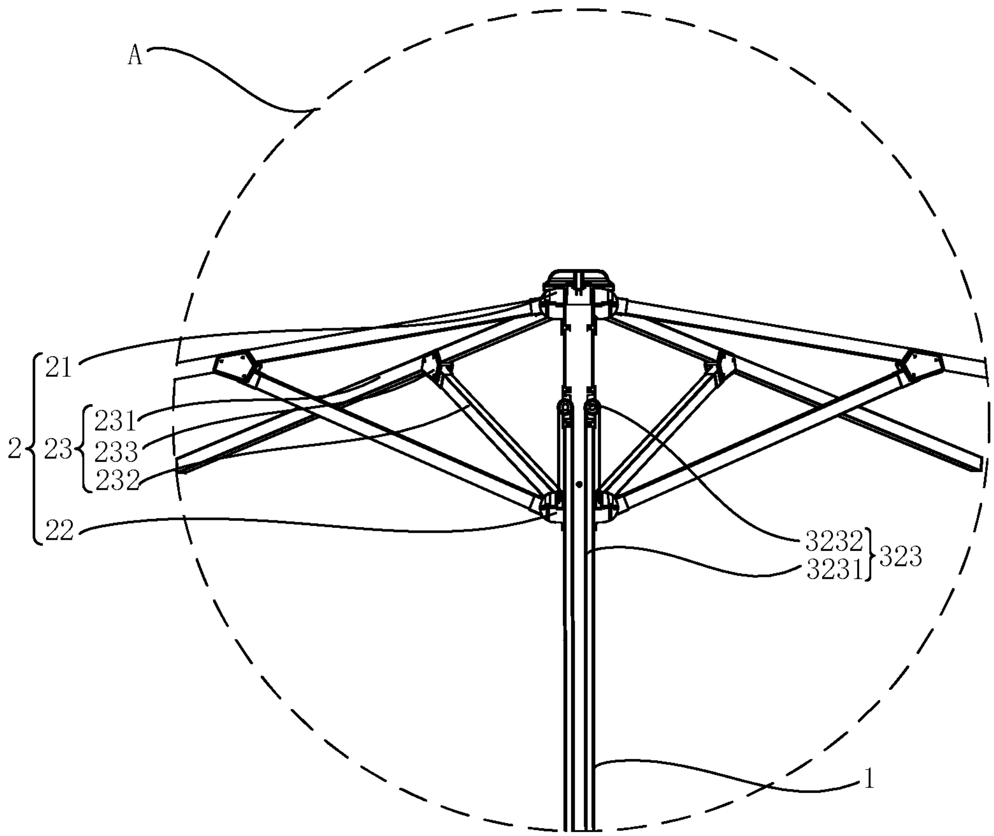

FIG. 3

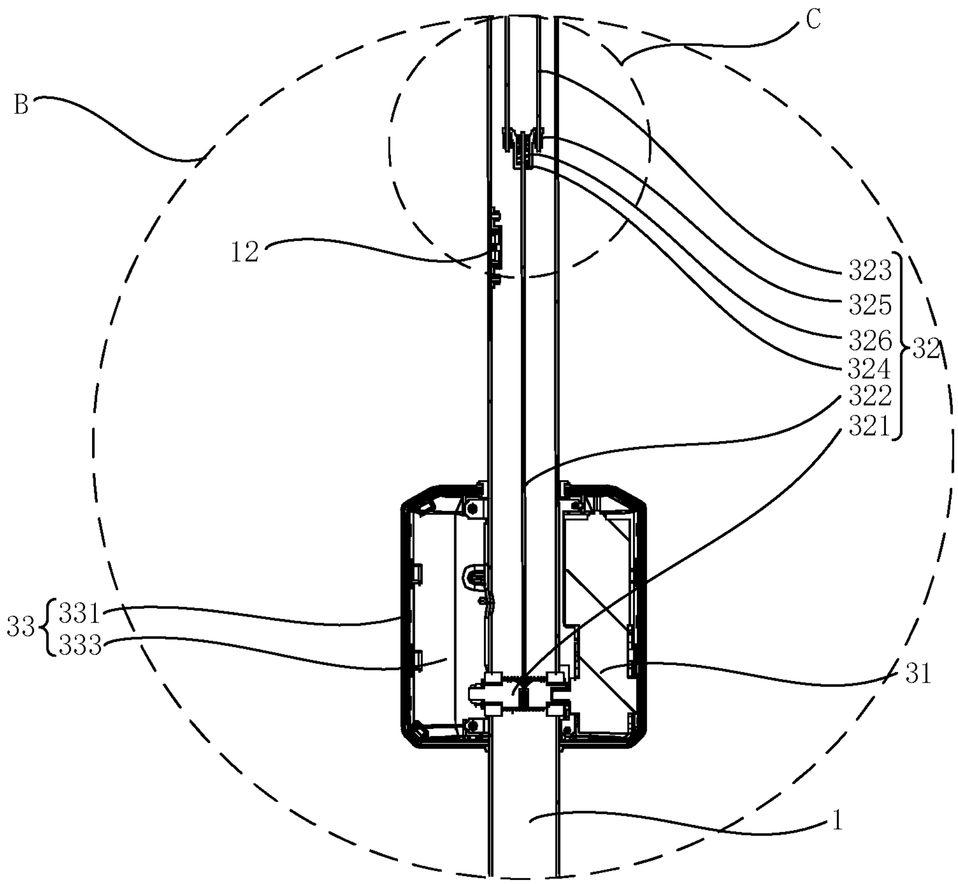

FIG. 4

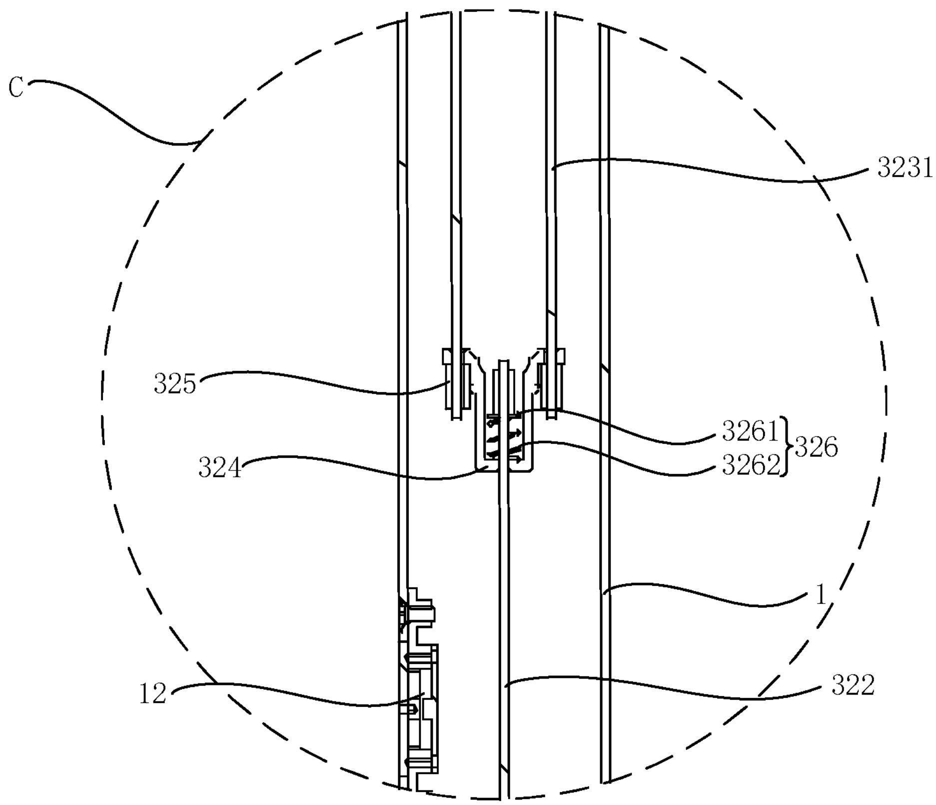

FIG. 5

LIFTING DEVICE AND SHADING DEVICE FOR SUNSHADES

CROSS-REFERENCE TO RELATED APPLICATIONS

The application claims priority to Chinese patent application No. 2022225095104, filed on Sep. 21, 2022, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

The present utility model belongs to the field of sunshade technology, more specifically, relates to a lifting device and shading device for sunshades.

BACKGROUND

As the social and economic levels rise, the number of people using their free time for outdoor activities is gradually increasing. As a result, sunshades have emerged on the market for outdoor shading and rain protection for users. Sunshades have gradually become an essential shading product for courtyard entertainment, sports, and leisure at home and abroad. In order to provide better shading effect for users, the sunshade will try to increase the area of the sunshade cloth, thereby increasing its shading area. However, it will result in a larger overall volume, heavier weight, and higher umbrella rod, which will cause inconvenience to users in actual use.

In the existing technology, for the convenience of users, a joystick mechanism is generally added to the umbrella rod for opening or closing the sunshade. Users manually rotate the joystick to drive the rotation of the winding shaft to loosen the umbrella rope wrapped on the winding shaft or wrap the umbrella rope around the winding shaft, thereby controlling the lifting or lowering of the umbrella frame and opening or closing the sunshade; However, this method is relatively labor-intensive for users in actual use, and it is easy to cause injuries due to the user's body being touched by the umbrella frame during operation. Users need to squat under the sunshade to operate and constantly observe the position of the umbrella frame, which causes inconvenience for users.

SUMMARY

The purpose of the present utility model is to provide a lifting device for sunshades, in order to solve the technical problems existing in the prior art that users have difficulty using sunshades and are prone to being injured by umbrella frames during use, which causes inconvenience to users.

To achieve the above objective, the technical solution adopted by the present utility model is to provide a lifting device for a sunshade, wherein the sunshade comprises a mounting frame, a sunshade cloth, a support mechanism and a lifting device set on the mounting frame, wherein the support mechanism comprises a fixed support member arranged at the top of the mounting frame, and a movable support member sliding on the mounting frame, and a support component connected to the fixed support member and the movable support member respectively and used for unfolding or closing the sunshade cloth, characterized in that, the lifting device comprises a driving motor, a transmission mechanism connected to the driving motor and used to drive the lifting motion of the movable support member relative to the mounting bracket, and a control component used to control the start and stop of the driving motor.

In one embodiment, the transmission mechanism comprises a winding shaft fixedly connected to the driving motor, a transmission component for driving the lifting motion of the movable support member relative to the mounting bracket, and a main wire wound on the winding shaft for connecting the transmission component.

In one embodiment, the transmission component comprises a sub wire and a pulley set on the mounting bracket and above the movable support member, wherein one end of the sub wire is connected to the main wire, and the other end bypasses the pulley and is connected to the movable support member.

In one embodiment, the number of transmission components is multiple, and the multiple transmission components are uniformly distributed along the circumference of the movable support member, and the transmission mechanism further comprises a splitter for connecting the main wire and the sub wire, and the splitter is provided with a through hole for the main wire and the sub wire to penetrate, the end of the main wire and the end of the sub wire are both equipped with wiring sleeves that are connected to the splitter.

In one embodiment, the transmission mechanism further comprises a buffer component arranged between the wiring sleeve and the splitter, wherein the buffer component comprises a support plate that is connected to the wiring sleeve and an elastic element arranged between the splitter and the support plate.

In one embodiment, the lifting device further comprises an upper limiter and a lower limiter set on the mounting bracket and electrically connected to the driving motor, used to limit the sliding range of the movable support member on the mounting bracket.

In one embodiment, the upper limiter and/or the lower limiter are limit protrusions on the mounting bracket and used to limit the movement of the movable support member, and the driving motor is equipped with an overload monitoring module.

In one embodiment, the upper limiter and/or the lower limiter are position sensors.

In one embodiment, the control component comprises a control housing, a battery detachably connected to the control housing, and a function key set on the control housing and electrically connected to the driving motor.

The beneficial effect of the lifting device provided by the present utility model for sunshades is that, compared with the prior art, the lifting device of the present utility model is used for sunshades, which comprises a mounting frame, a sunshade cloth, a support mechanism, and a lifting device set on the mounting frame. The support mechanism comprises a fixed support member set on the top of the mounting frame and a movable support member sliding on the mounting frame, and support components that are respectively connected to fixed and movable support members and used for unfolding or closing the sunshade cloth. The sunshade is firmly supported on the ground through the mounting frame, the sunshade cloth is fixedly connected to the support components, and the support components are respectively connected to the fixed support component set at the top of the mounting frame and the movable support component sliding on the mounting frame, and the movable support member is used to lift and move relative to the mounting frame, to realize the control of the unfolding or closing of the sunshade cloth; The lifting device comprises a driving motor, a transmission mechanism connected to the driving motor and used to drive the lifting motion of the movable support member relative to the mounting frame, and a control component used to control the start and stop of the driving motor. The driving motor and the movable support member are connected through the transmission mechanism to achieve the lifting motion of the movable support member relative to the mounting frame using the driving motor, thereby controlling the unfolding or closing of the sunshade cloth; Users can start and stop the driving motor by operating the control component, so as to drive the movable support member to lift and lower relative to the mounting frame, and thus control the unfolding or closing of the sunshade cloth. The use of driving method by the driving motor instead of manual driving can save labor and effectively prevent users from being injured by the sunshade during use, providing convenience for users and improving their experience.

The purpose of the present utility model is to provide a shading device for sunshades, in order to solve the technical problems existing in the prior art that users have difficulty using sunshades and are prone to being injured by umbrella frames during use, which causes inconvenience to users.

To achieve the above objective, the technical solution adopted by the present utility model is to provide a shading device, comprising a mounting frame, sunshade cloth, support mechanism, and a lifting device set on the mounting frame; The support mechanism comprises a fixed support member arranged at the top of the mounting frame, a movable support member sliding on the mounting frame, and a support component connected to the fixed support member and the movable support member respectively for unfolding or closing the sunshade cloth.

In one embodiment, the supporting component comprises a main support rod fixedly connected to the sunshade cloth, an auxiliary support rod fixedly connected to the movable support member, and a connecting bracket which is arranged at the end of the auxiliary support rod far from the movable support member, and the connecting bracket is slidably connected to the main support rod.

The beneficial effect of the shading device provided by the present utility model is that, compared with the prior art, the shading device of the present utility model comprises a mounting frame, a sunshade cloth, a support mechanism, and the lifting device set on the mounting frame. The support mechanism comprises a fixed support member set on the top of the mounting frame and a movable support member sliding on the mounting frame, and support components that are respectively connected to fixed and movable support members and used for unfolding or closing the sunshade cloth. The shading device is firmly supported on the ground through the mounting frame, the sunshade cloth is fixedly connected to the support components, and the support components are respectively connected to the fixed support component set at the top of the mounting frame and the movable support component sliding on the mounting frame, and the movable support member is used to lift and move relative to the mounting frame, to realize the control of the unfolding or closing of the sunshade cloth; The lifting device comprises a driving motor, a transmission mechanism connected to the driving motor and used to drive the lifting motion of the movable support member relative to the mounting frame, and a control component used to control the start and stop of the driving motor. The driving motor and the movable support member are connected through the transmission mechanism to achieve the lifting motion of the movable support member relative to the mounting frame using the driving motor, thereby controlling the unfolding or closing of the sunshade cloth; Users can start and stop the driving motor by operating the control component, so as to drive the movable support member to lift and lower relative to the mounting frame, and thus control the unfolding or closing of the sunshade cloth. The use of driving method by the driving motor instead of manual driving can save labor and effectively prevent users from being injured by the shading device during use, providing convenience for users and improving their experience.

BRIEF DESCRIPTION OF DRAWINGS

To more clearly illustrate the technical solutions in the embodiments of the present utility model, a brief description of the drawings required to be used in the embodiments or in the description of the prior art is presented below. It is obvious that the drawings described below are merely embodiments of present utility model, and that other drawings may be obtained from them without any creative effort on the part of the ordinary technician in the field.

FIG. **1** shows a schematic diagram of the three-dimensional structure of the shading device in the embodiment of the present utility model;

FIG. **2** shows a schematic diagram of the cross-sectional structure of the shading device in the embodiment of the present utility model;

FIG. **3** shows a schematic diagram of the enlarged structure at position A in FIG. **2**;

FIG. **4** shows a schematic diagram of the enlarged structure at position B in FIG. **2**;

FIG. **5** shows a schematic diagram of the enlarged structure at position C in FIG. **4**.

Wherein, each of the accompanying markings in Figures are:

1. Mounting frame; **11**. Upper limiter; **12**. Lower limiter; **2**. Support mechanism; **21**. Fixed support member; **22**. Movable support member; **23**. Support components; **231**. Main support rod; **232**. Auxiliary support rod; **233**. Connecting bracket; **3**. Lifting device; **31**. Driving motor; **32**. Transmission mechanism; **321**. Winding shaft; **322**. Main wire; **323**. Transmission components; **3231**. Sub wire; **3232**. Pulley; **324**. Splitter; **325**. Wiring sleeve; **326**. Buffer component; **3261**. Support plate; **3262**. Elastic element; **33**. Control component; **331**. Control housing; **332**. Function key; **333**. Battery.

DETAILED DESCRIPTION OF THE EMBODIMENTS

In order to make the technical problems, technical solutions and beneficial effects to be solved by present utility model more clearly understood, the following is a further detailed description of the present utility model in conjunction with the accompanying drawings and implementation examples. It should be understood that the specific embodiments described herein are only intended to explain the present utility model and are not intended to limit it.

It should be noted that when a component is referred to as “fixed to” or “set on” another component, it may be directly or indirectly on that other component. When a component is referred to as “connected to” another component, it may be directly connected to another component or indirectly connected to that other component.

It should be understood that the terms “length”, “width”, “up”, “down”, “front”, “back”, “left”, “right”, “vertical”, “horizontal”, “top”, “bottom”, “inside”, “outside”, etc. indicate an orientation or position relationship based on the orientation or position relationship shown in the attached drawings, which are intended only to facilitate and simplify the description of the present utility model, not to indicate or imply that the device or component referred to must have a particular orientation, be constructed and operated in a particular orientation, and therefore cannot be construed as a limitation on the present utility model.

In addition, the terms "first" and "second" are used for descriptive purposes only and are not to be construed as indicating or implying relative importance or implicitly specifying the quantity of technical features indicated. Thus, the features qualified with "first" and "second" may explicitly or implicitly include one or more of these features. In the description of the present utility model, "multiple" refers to two or more, unless otherwise specified.

Please refer to FIGS. **1** to **2** together. The embodiment of the present utility model provides a lifting device **3**, which is used for a sunshade. The sunshade comprises a mounting frame **1**, a sunshade cloth, a support mechanism **2**, and a lifting device **3** set on the mounting frame **1**. The support mechanism **2** comprises a fixed support member **21** set on the top of the mounting frame **1** and a movable support member **22** sliding on the mounting frame **1**, and support components **23** that are respectively connected to fixed support member **21** and movable support member **22** and used for unfolding or closing the sunshade cloth. The sunshade is firmly supported on the ground through the mounting frame **1**, the sunshade cloth is fixedly connected to the support components **23**, and the support components **23** are respectively connected to the fixed support component **21** set at the top of the mounting frame **1** and the movable support component **22** sliding on the mounting frame **1**, and the movable support member **22** is used to lift and move relative to the mounting frame **1**, to realize the control of the unfolding or closing of the sunshade cloth; The lifting device **3** comprises a driving motor **31**, a transmission mechanism **32** connected to the driving motor **31** and used to drive the lifting motion of the movable support member **22** relative to the mounting frame, and a control component **33** used to control the start and stop of the driving motor **31**. The driving motor **31** and the movable support member **22** are connected through the transmission mechanism **32** to achieve the lifting motion of the movable support member **22** relative to the mounting frame **1** using the driving motor **31**, thereby controlling the unfolding or closing of the sunshade cloth; Users can start and stop the driving motor **31** by operating the control component **33**, so as to drive the movable support member **22** to lift and lower relative to the mounting frame **1**, and thus control the unfolding or closing of the sunshade cloth. The use of driving method by the driving motor **31** instead of manual driving can save labor and effectively prevent users from being injured by the sunshade during use, providing convenience for users and improving their experience.

Please refer to FIGS. **3** to **5** together. As a specific implementation of the lifting device **3** for the sunshade in the embodiment of the present utility model, a fixed bracket for installing the lifting device **3** is set on the mounting bracket **1** of the sunshade, and the driving motor **31** and control component **33** are stably installed on the mounting bracket **1** using the fixed bracket; There is no restriction on the connection method between the fixed bracket and the mounting bracket **1** here; Optionally, the fixed bracket can be installed on the mounting bracket **1** using detachable or non detachable connections; Optionally, the fixed bracket can be installed on the mounting bracket **1** through threaded connection, which is convenient for assembly and disassembly, maintenance or replacement of accessories by operators, and helps to reduce the cost of using the sunshade.

Optionally, the transmission mechanism **32** of the lifting device **3** comprises a winding shaft **321** fixedly connected to the driving motor **31**, a transmission component **323** for driving the lifting motion of the movable support member **22** relative to the mounting frame **1**, and a main wire **322** wound on the winding shaft **321** and connected to the transmission component **323**. The winding shaft **321** is rotatably set on the mounting frame **1** and fixedly connected to the output shaft of the driving motor **31**. The driving motor **31** is used to drive the winding shaft **321** to rotate, loosen the main wire **322** wound on the winding shaft **321** or wind the main wire **322** on the winding shaft **321**, thereby driving the transmission component **323** connected to the main wire **322**. Then, the transmission component **323** is used to drive the movable support member **22** to move up and down relative to the mounting frame **1**, to control the unfolding or closing of the sunshade cloth. The use of the driving motor **31** to drive the sunshade makes it more labor-saving for users.

Optionally, the transmission component **323** comprises a sub wire **3231** and a pulley **3232** located on the mounting frame **1** and above the movable support member **22**. One end of the sub wire **3231** is connected to the main wire **322**, while the other end bypasses the pulley **3232** and is connected to the movable support member **22**. The main wire **322** and the movable support member **22** are connected through the sub wire **3231**, and the pulley **3232** set above the movable support member **22** is used as the support point. When the driving motor **31** drives the winding shaft **321** to rotate and releases the main wire **322** wrapped on the winding shaft **321**, the connection point between the main wire **322** and the sub wire **3231** moves upwards relative to the mounting frame **1**. The sub wire **3231** moves relative to the pulley **3232** and causes the movable support member **22** connected to the other end of the sub wire **3231** to slide downwards relative to the mounting frame **1**, thereby increasing the distance between the fixed support member **21** and the movable support member **22**, and thus closing the sunshade cloth; When the driving motor **31** drives the winding shaft **321** to rotate and winds the main wire **322** onto the winding shaft **321**, the connection point between the main wire **322** and the sub wire **3231** moves downwards relative to the mounting frame **1**, and the sub wire **3231** moves relative to the pulley **3232**, causing the movable support member **22** connected to the other end of the sub wire **3231** to slide upwards relative to the mounting frame **1**, thereby reducing the distance between the fixed support member **21** and the movable support member **22**, and thus unfolding the sunshade cloth.

Optionally, the number of transmission components **323** is multiple, and the multiple transmission components **323** are evenly distributed along the circumference of the movable support member **22**, making the force on the movable support member **22** more uniform when the transmission component **323** pulls the movable support member **22**. This ensures that the movable support member **22** moves up and down relative to the mounting frame **1** in a direction perpendicular to the horizontal plane, which is beneficial for reducing the friction force between the movable support member **22** and the outer wall of the mounting frame **1** during the sliding process, making the sunshade more stable during the unfolding or closing process, which is beneficial for improving the user experience. Optionally, the number of transmission components **323** is two, and the two transmission components **323** are symmetrically set on opposite sides of the mounting frame **1**, to ensure that the movable support member **22** is evenly stressed during sliding relative to the mounting frame **1**, while effectively controlling the production cost of the sunshade.

Optionally, the transmission mechanism **32** further comprises a splitter **324** for connecting the main wire **322** and the sub wire **3231**. The splitter **324** is provided with a through hole for the main wire **322** and the sub wire **3231** to penetrate. The end of the main wire **322** and the end of the sub wire **3231** are both provided with a wiring sleeve **325** that is connected to the splitter **324**. By setting a wiring sleeve **325** at the end of the main wire **322** and the end of the sub wire **3231**, the main wire **322** and the sub wire **3231** are threaded through the through-hole on the splitter **324**, and the wiring sleeve **325** is connected to the splitter **324**, thus achieving the transmission of the power output from the driving motor **31** to multiple sub wires **3231** through the main wire **322**. By using multiple sub wires **3231** to pull the movable support member **22** relative to the mounting frame **1**, the sunshade cloth can be controlled to unfold or close.

Optionally, the transmission mechanism **32** further comprises a buffer component **326** arranged between the wiring sleeve **325** and the splitter **324**. The buffer component **326** is used to buffer the impact force when the driving motor **31** starts, which can protect the driving motor **31** and the transmission mechanism **32**, and is conducive to extending the service life of the sunshade; Optionally, the buffer component **326** comprises a support plate **3261** that is connected to the wiring sleeve **325** and an elastic member **3262** that is arranged between the splitter **324** and the support plate **3261**. The support plate **3261** that is set on the splitter **324** is connected and fixed with the wiring sleeve **325** to prevent the main wire **322** and/or sub wire **3231** from detaching from the separator under external force, and the elastic member **3262** is used to buffer the impact force when the driving motor **31** starts, to protect the driving motor **31** and transmission mechanism **32**. Optionally, the elastic element **3262** may be a spring or a shrapnel.

Please refer to FIGS. **1** and **5** together. In this embodiment, the lifting device **3** further comprises an upper limiter **11** and a lower limiter **12** set on the mounting frame **1** and electrically connected to the driving motor **31**, which are used to limit the sliding range of the movable support member **22** on the mounting frame **1**. When the user needs to use the sunshade, the control component **33** is used to start the lifting device **3**. The driving motor **31** drives the movable support member **22** to slide relative to the mounting frame **1**, and when the movable support member **22** reaches the upper limiter **11**, the lifting device **3** stops operating. The driving motor **31** uses its own locking force to tighten the main wire **322**, making the sunshade cloth in a unfolded state; When the user needs to close the sunshade, the control component **33** is used to start the lifting device **3**, the driving motor **31** to release its locking force and drive the winding shaft **321** to rotate in the opposite direction. The main wire **322** is wound around the winding shaft **321**, and the movable support member **22** slides downwards relative to the mounting frame **1** under the gravity of the sunshade cloth and support mechanism **2**, as well as the driving effect of the driving motor **31**. When the movable support member **22** reaches the lower limiter **12**, the lifting device **3** stops operating, and the driving motor **31** uses its own locking force to tighten the main wire **322**, causing the sunshade cloth to be in a closed state. The specific structures of upper limiter **11** and lower limiter **12** on mounting bracket **1** are not limited here. Optionally, the upper limiter **11** and/or lower limiter **12** are position sensors. When the movable support member **22** slides to the position of the upper limiter **11** or lower limiter **12** relative to the mounting frame **1**, the driving motor **31** stops operating and uses its own locking force to tighten the main wire **322**, keeping the sunshade cloth in an unfolded or closed state. Optionally, the upper limiter **11** and/or the lower limiter **12** are limit protrusions protruding from the mounting bracket **1** and used to limit the movement of the movable support member **22**. An overload monitoring module is installed on the driving motor **31**. When the movable support member **22** slides relative to the mounting bracket **1** to come into contact with the upper limiter **11** or the lower limiter **12**, the movable support member **22** stops sliding under the limit action of the upper limiter **11** or the lower limiter **12**, resulting in an increase in the resistance of the driving motor **31** to drive the winding shaft **321**. At this point, the overload monitoring module of driving motor **31** detects that the load of driving motor **31** exceeds the limit and controls it to stop operating. At the same time, it uses its own locking force to tighten the main wire **322**, keeping the sunshade cloth in an unfolded or closed state. It is characterized by simple structure and easy production.

Please refer further to FIGS. **1** and **4**. In this embodiment, the control component **33** of the lifting device **3** comprises a control housing **331**, a battery **333** detachably connected to the control housing **331**, and a function key **332** electrically connected to the driving motor **31**, which is set on the control housing **331**. The control housing **331** is firmly installed on the mounting frame **1** through a fixed bracket set on the mounting frame **1**, and a mounting portion for detachably connecting the battery **333** is set on the control housing **331**, By installing a large capacity battery **333** on the mounting portion of the control housing **331**, the required electrical energy is provided for the operation of the lifting device **3**. While meeting the user's usage needs, the charging frequency is minimized as much as possible. In special circumstances, the battery **333** can be removed to charge the user's mobile terminal, which is beneficial for improving the user experience. Optionally, the function key **332** electrically connected to the driving motor **31** on the control housing **331** comprises an up key for indicating the upward sliding of the movable support member **22** relative to the mounting frame **1** and a down key for indicating the downward sliding of the movable support member **22** relative to the mounting frame **1**. The user only needs to press the function key **332** to achieve the lifting motion of the movable support member **22** relative to the mounting frame **1**, thereby controlling the unfolding or closing of the sunshade cloth. The use of driving method by the driving motor **31** instead of manual driving can save labor and effectively prevent users from being injured by the sunshade during use, providing convenience for users and improving their experience. Optionally, a signal receiving terminal is set on the control panel inside the control housing **331**. Users can send a signal to the signal receiving terminal through the remote control to remotely start the lifting device **3**, and drive the movable support member **22** to move relative to the mounting frame **1**, thereby controlling the unfolding or closing of the sunshade cloth. It is characterized by simple structure, labor-saving and convenient operation, providing convenience for users and improving user experience.

Please refer to FIGS. **1** to **3** together. The embodiment of the present utility model provides a shading device, which comprises a mounting frame **1**, a sunshade cloth, a support mechanism **2**, and the lifting device **3** set on the mounting frame **1**. The support mechanism **2** comprises a fixed support member **21** set on the top of the mounting frame **1** and a movable support member **22** sliding on the mounting frame **1**, and the support components **23** that are respectively connected to fixed support member **21** and movable support member **22** and used for unfolding or closing the sunshade cloth. The shading device is firmly supported on the ground through the mounting frame **1**, the sunshade cloth is fixedly connected to the support components **23**, and the support components **23** are respectively connected to the fixed support component **21** set at the top of the mounting frame **1** and the movable support component **22** sliding on the mounting frame **1**, and the movable support member **22** is used to lift and move relative to the mounting frame **1**, to realize the control of the unfolding or closing of the sunshade cloth; The lifting device **3** comprises a driving motor **31**, a transmission mechanism **32** connected to the driving motor **31** and used to drive the lifting motion of the movable support member **22** relative to the mounting frame, and a control component **33** used to control the start and stop of the driving motor **31**. The driving motor **31** and the movable support member **22** are connected through the transmission mechanism **32** to achieve the lifting motion of the movable support member **22** relative to the mounting frame **1** using the driving motor **31**, thereby controlling the unfolding or closing of the sunshade cloth; Users can start and stop the driving motor **31** by operating the control component **33**, so as to drive the movable support member **22** to lift and lower relative to the mounting frame **1**, and thus control the unfolding or closing of the sunshade cloth. The use of driving method by the driving motor **31** instead of manual driving can save labor and effectively prevent users from being injured by the sunshade umbrella during use, providing convenience for users and improving their experience.

Optionally, the support component **23** of the support mechanism **2** comprises a main support rod **231** fixedly connected to the sunshade cloth and the fixed support member **21**, an auxiliary support rod **232** fixedly connected to the movable support member **22**, and a connecting bracket **233**. The connecting bracket **233** is arranged at the end of the auxiliary support rod **232** away from the movable support member **22**, and the connecting bracket **233** is slidably connected to the main support rod **231**. When the movable support member **22** is driven by the lifting device **3** to move up and down relative to the mounting frame **1**, the auxiliary support rod **232** fixedly connected to the movable support member **22** also moves up and down relative to the mounting frame **1**, so that the connecting bracket **233** set at the end of the auxiliary support rod **232** away from the movable support member **22** slides relative to the main support rod **231**, and the distance between the movable support member **22** and the fixed support member **21** changes accordingly, thereby achieving the control of the unfolding or closing of the sunshade cloth.

The above is only a preferred embodiment of this embodiment and is not intended to limit this embodiment. Any modifications, equivalent replacements and improvements made within the spirit and principles of this embodiment shall be included within the scope of protection of this embodiment.

What is claimed is:

1. A lifting device for a sunshade, characterized in that the sunshade comprises a mounting frame, a sunshade cloth, a support mechanism and a lifting device set on the mounting frame, wherein the support mechanism comprises a fixed support member arranged at the top of the mounting frame, and a movable support member sliding on the mounting frame, and a support component connected to the fixed support member and the movable support member respectively and used for unfolding or closing the sunshade cloth, the lifting device comprises a driving motor, a transmission mechanism connected to the driving motor and used to drive the lifting motion of the movable support member relative to the mounting bracket, and a control component used to control the start and stop of the driving motor;

wherein the transmission mechanism comprises a winding shaft fixedly connected to the driving motor, a transmission component for driving the lifting motion of the movable support member relative to the mounting bracket, and a main wire wound on the winding shaft for connecting the transmission component;

wherein the transmission component comprises a sub wire and a pulley set on the mounting bracket and above the movable support member, one end of the sub wire is connected to the main wire, and the other end bypasses the pulley and is connected to the movable support member;

wherein the number of transmission components is multiple, and the multiple transmission components are uniformly distributed along the circumference of the movable support member, and the transmission mechanism further comprises a splitter for connecting the main wire and the sub wire, and the splitter is provided with a through hole for the main wire and the sub wire to penetrate, the end of the main wire and the end of the sub wire are both equipped with wiring sleeves that are connected to the splitter; and

wherein the transmission mechanism further comprises a buffer component arranged between the wiring sleeve and the splitter, and the buffer component comprises a support plate that is connected to the wiring sleeve and an elastic element arranged between the splitter and the support plate.

2. A lifting device for a sunshade in claim **1**, characterized in that the lifting device further comprises an upper limiter and a lower limiter set on the mounting bracket and electrically connected to the driving motor, used to limit the sliding range of the movable support member on the mounting bracket.

3. The lifting device for a sunshade of claim **2**, characterized in that the upper limiter and/or the lower limiter are limit protrusions on the mounting bracket and used to limit the movement of the movable support member, and the driving motor is equipped with an overload monitoring module.

4. The lifting device for a sunshade of claim **2**, characterized in that the upper limiter and/or lower limiter are position sensors.

5. The lifting device for a sunshade in claim **1**, characterized in that the control component comprises a control housing, a battery detachably connected to the control housing, and a function key set on the control housing and electrically connected to the driving motor.

6. A shading device, characterized in that it comprises a mounting frame, a sunshade cloth, and the lifting device according to claim **1** set on the mounting frame.

7. The shading device of claim **6**, characterized in that the supporting component comprises a main support rod fixedly connected to the sunshade cloth, an auxiliary support rod fixedly connected to the movable support member, and a connecting bracket which is arranged at the end of the auxiliary support rod far from the movable support member, and the connecting bracket is slidably connected to the main support rod.

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